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Knapp et al.

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(54) **STRAWBERRY PLANT NAMED 'UCD MOXIE'**

(50) Latin Name: ***Fragaria* x *ananassa Duchesne***
Varietal Denomination: **UCD Moxie**

(71) Applicant: **The Regents of the University of California**, Oakland, CA (US)

(72) Inventors: **Steven J. Knapp**, Davis, CA (US); **Glenn S. Cole**, Davis, CA (US); **Douglas V. Shaw**, Davis, CA (US); **Kirk D. Larson**, Santa Ana, CA (US)

(73) Assignee: **The Regents of the University of California**, Oakland, CA (US)

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Related U.S. Application Data

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(52) **U.S. Cl.**
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(58) **Field of Classification Search**
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Primary Examiner — Anne Marie Grunberg
(74) *Attorney, Agent, or Firm* — Kilpatrick Townsend & Stockton LLP

(57) **ABSTRACT**

'UCD Moxie' is a day-neutral cultivar of a strawberry plant that provides high yields, has resistance to Fusarium wilt and Verticillium wilt, and produces fewer runners than many high-yielding day-neutral cultivars.

7 Drawing Sheets

Genus and species: The strawberry cultivar of this invention is botanically identified as *Fragaria* x *ananassa* Duchesne.

Variety denomination: The variety denomination is 'UCD Moxie'.

BACKGROUND OF THE INVENTION

This invention relates to a new and distinct day-neutral strawberry cultivar designated as 'UCD Moxie', which originated from a cross performed in the winter of 2011 between parent 08C123P001 ('UCD Royal Royce', U.S. plant patent application Ser. No. 16/501,374) and proprietary germplasm parent 07C092P003 (unpatented). Seeds of the cross were harvested from greenhouse-grown plants in the spring of 2011 and germinated in June 2011. Seedlings were transplanted to a greenhouse in July 2011 and transplanted to the field in October 2011. 'UCD Moxie' was selected and clones were first harvested in 2012. 'UCD Moxie' has been asexually propagated since 2012.

The plant of this selection was originally designated '11C141P001' (also represented as 11.141-1) and later called '16DN012' or 'UC12' for evaluation in field trials. 'UCD Moxie' was also called '15 MBA-4' for certain testing trials.

BRIEF SUMMARY OF THE INVENTION

'UCD Moxie' is a day-neutral (ever-bearing) strawberry cultivar selected for increased marketable fruit yield, resis-

tance to Fusarium wilt and Verticillium wilt, fruit firmness, extended shelf-life, and decreased stolon production. 'UCD Moxie' consistently produces more marketable fruit per hectare compared to 'Monterey' (U.S. Plant Pat. No. 19,767) or 'Cabrillo' (U.S. Plant Pat. No. 27,830); and has fewer stolons (runners) per plant than 'Monterey' or 'Cabrillo'. Cumulative marketable fruit yield of 'UCD Moxie' over a full growing season surpasses that of 'UCD Royal Royce' (U.S. plant patent application Ser. No. 16/501,374). 'UCD Moxie' also has superior resistance to Fusarium wilt and Verticillium wilt compared to 'UCD Royal Royce' and 'UCD Valiant' (U.S. plant patent application Ser. No. 16/501,375); and has superior resistance to Fusarium wilt compared to 'Cabrillo', 'Monterey', and 'Albion' (U.S. Plant Pat. No. 16,228). 'UCD Moxie' exhibits increased Fusarium resistance compared to parent 08C123P001, and exhibits earlier flowering and increased numbers of stolons compared to parent 07C09P003

'UCD Moxie' was genotyped with a 35,000-SNP array (Hardigan et. al., *Plant Genome* 11:180049, 2018). The variety has a unique DNA profile compared to 'Cabrillo', 'Monterey', 'UCD Royal Royce', and 'UCD Valiant'. The pairwise genetic distances between 'UCD Moxie' and 'UCD Valiant', 'UCD Royal Royce', 'Cabrillo', and 'Monterey' are estimated to be 0.223, 0.167, 0.296, and 0.287, respectively based on the 35,000-SNP genotype analysis. 'UCD Moxie' also has a unique DNA profile compared to each of its parents, proprietary germplasm varieties 08C123P001 and 07C092P003.

'UCD Moxie' is maintained by annual asexual propagation by stolons in Winters, Calif.

BRIEF DESCRIPTION OF THE DRAWINGS

The colors in the photograph are depicted as nearly true as is reasonably possible to obtain in color reproductions of this type.

FIG. **1** shows fruit of 'UCD Moxie' in cross-section.

FIG. **2** illustrates color of the fruit.

FIG. **3** depicts general flowering and fruiting characteristics of 'UCD Moxie'.

FIG. **4** shows representative fruit trusses of the plant.

FIG. **5** shows representative flowers of the plant.

FIG. **6** shows representative leaves of the plant.

FIG. **7** depicts plant habit.

DETAILED DESCRIPTION OF THE INVENTION

Fruit Production

'UCD Moxie' and comparison cultivars were asexually propagated in high-elevation nurseries in Dorris and Mcdoel, Calif. for field testing in 2015-2016, 2016-2017, and 2017-2018. Clones were harvested according to commercial planting schedules, which were calibrated by the number of chill hours required for optimum production in Santa Maria and Prunedale, Calif.

Cultivar testing in small-plot yield trials was performed as follows. Test day-neutral cultivars and comparison cultivars were field tested in Oso Flaco (Santa Maria, Calif.) and Prunedale, Calif. in 2015-2016. Fruit was harvested once or twice per week over the spring and summer growing season: March 16 to Aug. 18, 2016 in Santa Maria (31 harvests) and April 2 to Aug. 27, 2016 in Prunedale (30 harvests). 'UCD Moxie' was selected on the basis of fruit appearance, size, shelf-life, and cumulative marketable fruit yield for a second year (2016-17) of replicated testing in Santa Maria and Prunedale (Tables 1-2). Fruit was harvested once or twice per week over the growing season: March 13 to Aug. 21, 2017 in Santa Maria (25 harvests) and April 3 to Sep. 1, 2017 in Prunedale (44 harvests in Prunedale). To highlight differences among cultivars, the yield data were displayed in three ways: (a) per plant yields for individual harvests for each location x year combination for 'UCD Moxie', 'Monterey', 'San Andreas', and 'Cabrillo'; (b) per plant cumulative marketable yields for 'UCD Moxie', 'Monterey', 'San Andreas', and 'Cabrillo'; and (c) per plant cumulative marketable yields tested in both years and locations. Entry ranks were highly consistent and the best and worst performing cultivars were virtually identical over locations and years.

Table 1 shows marketable yield (g/plant) for 'UCD Moxie' compared to that for 'Monterey', 'San Andreas' (U.S. Plant Pat. No. 19,975), and 'Cabrillo' tested in small-plot yield trials in Santa Maria, Calif. and Prunedale, Calif. in 2015-16 and 2016-17.

Table 2 shows the range in increase in yield compared to 'Monterey', 'San Andreas', and 'Cabrillo' for the small-plot yield trials.

TABLE 1

Least-square means for cumulative marketable yield tested in small-plot yield trials in Santa Maria, CA and Prunedale, CA in 2015-2016 and 2016-2017

	Santa Maria		Prunedale		Yield across Locations and Years
Cultivar	2015-16	2016-17	2015-16	2016_17	(g/plant)
UCD Moxie	1,815	1,460	1,627	2,580	2,033
Cabrillo	1,617	2,073	1,362	2,299	1,838
Monterey	1,115	1,324	1,077	1,722	1,310
San Andreas	1,229	1,096	1,048	1,569	1,236

TABLE 2

Least-square means for cumulative marketable yield (g/plant) across years and locations in the small-plot yield trials. The ranges for yield increases over comparison cultivars were estimated from least square means for individual environments (see, Table 1). (Percent Marketable Fruit = (Cumulative Marketable Fruit Yield)/(Cumulative Total Fruit Yield).

Cultivar	Yield (g/plant)	Percent Marketable Fruit	Yield Increase Range Over San Andreas	Yield Increase Range Over Monterey	Yield Increase Range Over Cabrillo
UCD Moxie	2,033	87-89	33-64%	10-63%	0-19%
Cabrillo	1,838	86-87			
Monterey	1,310	79-82			
San Andreas	1,236	76-84			

The cumulative marketable fruit yields of 'UCD Moxie' were significantly greater than 'Monterey' ($p<0.0001$) and 'San Andreas' ($p<0.0001$) across years and locations (Table 1), but not significantly greater than 'Cabrillo ($p=0.13$). The per plant yields of 'UCD Moxie' were 30% lower than 'Cabrillo' in Santa Maria in 2015-2016. Otherwise, the per plant yield of 'UCD Moxie' were 12 to 64% greater than 'Cabrillo', 'Monterey', and 'San Andreas' (Table 2). As confirmed in 2017-2018 large-plot yield trials, described below, 'UCD Moxie' yields increased into late summer and fall, eventually surpassing all other comparison cultivar yields. (The late season yield trends were evident in the small-plot yield trials in 2015-16, and 2016-17, but harvesting was stopped in August or September to prepare for planting the next cycle of trials.) The mean fruit size for 'UCD Moxie' (32.1 g/fruit) was significantly different from 'Cabrillo' (30.0 g/fruit; $p<0.054$) or Monterey (28.4 g/fruit; $p=0.001$), and 'San Andreas' (26.7 g/fruit; $p<0.0001$).

The marketable fruit percentages for 'UCD Moxie' and 'Cabrillo' were comparable (86-89%) and superior to 'Monterey' and 'San Andreas' (76 to 84%) (Table 2).

Significantly less runner production was observed in 'UCD Moxie' compared to the other high yielding cultivars tested in small-plot yield trials. Runner production differences among cultivars could not be absolutely quantified because growers trimmed runners to prevent the diversion of energy away from fruit production; however, it is estimated based on observation that runner production was reduced by at least 50% for 'UCD Moxie' compared to runner production by 'San Andreas', 'Monterey', and 'Cabrillo'. This was confirmed in large-plot yield trials described below.

'UCD Moxie' was selected, with other test varieties, for advanced testing in large-plot yield trials (150 plants/plot) with five commercial growers in 2017-18 (Table 3-5). The selection criteria were: high cumulative marketable yields, high marketable fruit percentages, resistance to bruising and deterioration with harvest, handling, and storage, fruit appearance, and reduced runner production in coastal production environments. The production systems for large-plot yield trials were diverse and included: low-input organic (Santa Maria, Calif.), low-input fumigated (Salinas, Calif.), and high-input fumigated (Oso Flaco, Calif., Prunedale, Calif. and Moss Landing, Calif.). Fruit was harvested from February 2 to October 5 in Oso Flaco (52 harvests), April 3 to October 4 in Santa Maria (53 harvests), April 12 to October 6 in Prunedale (48 harvests), May 7 to October 8 in Moss Landing (45 harvests), and March 9 to October 12 in Salinas (36 harvests). The residual for statistical analyses was the entry x location interaction mean square.

TABLE 3

Least square means for cumulative marketable fruit yield (cartons/acre) for 'UCD Moxie', 'Monterey', and 'Cabrillo' grown in unreplicated large-plot (150 plant) yield trials in five locations in 2017-18, where a carton = eight clamshells and eight clamshells/carton × one pound/clamshell = eight pounds/carton.

Cultivar	Salinas	Santa Maria	Oso Flaco	Prunedale	Mossing Landing	Across Locations
UCD Moxie	8,512	10,707	18,944	16,310	11,814	13,257
Cabrillo	5,201	6,892	17,569	14,057	9,465	10,637
Monterey	NA	7,891	14,731	11,940	10,257	10,274

TABLE 4

Least square means for fruit weight (g/fruit) for 'UCD Moxie', 'Monterey', and 'Cabrillo' grown in unreplicated large-plot (150 plant) yield trials in in five locations in 2017-18, where fruit weight = (weight of fruit per clamshell)/(number of fruit per clamshell).

Cultivar	Salinas	Santa Maria	Oso Flaco	Prunedale	Mossing Landing	Across Locations
UCD Moxie	27.9	22.8	30.2	30.2	34.8	27.2
Cabrillo	26.1	21.9	28.0	27.5	34.0	27.5
Monterey	NA	21.6	29.4	27.9	33.8	27.8

TABLE 5

Statistical significance (Pr > F) of differences between least square means for cumulative fruit yield (cartons/acres) and fruit weight (g/fruit) for 'UCD Moxie', 'Monterey', and 'Cabrillo' tested in unreplicated large-plot (150 plant) yield trials in five locations in 2017-18.

	Yield		Fruit Weight (g/fruit)	
Comparison	Difference (carton/acre)	Pr > F	Difference (g/fruit)	Pr > F
UCD Moxie-Monterey	2,983	0.0075	1.34	0.0156
UCD Moxie-Cabrillo	2,621	0.0106	1.67	0.0024

The mean cumulative marketable yields 'UCD Moxie' was 13,257 cartons/acre=106,056 pounds/acre across production systems and locations (Table 3). Reduced runner production was observed for 'UCD Moxie' across trials. 'UCD Moxie' yields surpassed all other cultivars late in the season; and was the highest yielding cultivar by cumulative marketable fruit yields across the season. 'UCD Moxie' produced 2,621 cartons/acre more than 'Monterey' (p=0.008) and 2,449 cartons/acre more than 'Cabrillo' (p=0.01) (Tables 3 & 5). 'UCD Moxie' fruit weights were significantly larger than 'Monterey' (p<0.02) and 'Cabrillo' (p<0.002) (Tables 4 & 5).

To assess the quality of freshly harvested fruit, firmness (grams force), total soluble solids (SS) concentration, and titratable acid (TA) concentrations were measured from samples of fruit harvested on three dates from each location in 2017-18 (Tables 6-7). Harvest dates were one month apart with one replication per harvest date, 10 sub-samples per replication for firmness, and three subsamples per replication for SS and TA. Firmness was quantified with a handheld penetrometer measuring the grams of force needed to puncture the fruit. SS and TA concentrations were quantified with benchtop instruments. The SS to TA ratio provides a relative measure of sweetness. To assess shelf-life, fruit weight (g/clamshell), SS, brightness (ordinal scale with 1=excellent to 5=unmarketable), liquid leakage (g/clamshell), and mold incidence (%) were quantified from samples of fruit harvested on two dates from each location with fruit stored under standard 4° C. conditions for 0, 7, 14, and 21 days (Table 8). Harvest dates were one month apart with one replication per harvest date.

TABLE 6

Least-square means (LSMs) for firmness, soluble solids concentration (SS), and titratable acid concentration (TA) for 'UCD Moxie', 'Monterey', and 'Cabrillo'grown in five locations in 2017-18. LSMs were estimated from three harvest dates per location, one biological replication per harvest date, 10 sub-samples per harvest date for firmness, and three sub-samples per harvest date for SS and TA.

Cultivar	Firmness (g force)	SS (%)	Titrable Acids (g/100 ml)	SS/TA
UCD Moxie	425.15	7.58	0.74	10.38
Cabrillo	359.61	8.05	0.77	10.50
Monterey	294.55	8.71	0.77	11.48

TABLE 7

Statistical significance (Pr > F) of differences between least square means for SS, TA, and SS/TA for 'UCD Moxie', 'Monterey', and 'Cabrillo' tested in unreplicated large-plot yield trials in five locations 2017-18.

	Soluble Solids Concentration (SS)		Titratable Acids Concentration (TA)		SS/TA	
Comparison	Least Square Mean Difference (%)	Pr > F	Least Square Mean Difference (g/100 ml)	Pr > F	Least Square Mean Differ-ence	Pr > F
UCD Moxie-Monterey	−1.13	0.0049	−0.03	0.3250	−1.09	0.0160
UCD Moxie-Cabrillo	−0.47	0.2475	−0.04	0.2082	−0.12	0.7995

TABLE 8

Least-square means (LSMs) for fruit weight (g/clamshell), soluble solids concentration (SS), fruit brightness, liquid leakage, and mold formation for 'UCD Moxie', 'Monterey', and 'Cabrillo' grown in four locations in 2017-18 and stored for zero to 21 days postharvest. LSMs were estimated from two harvest dates per location

Cultivar	Days Post-Harvest Storage	Weight (g/clamshell)	Soluble Solids (%)	Brightness	Liquid Leakage (g)	Mold (%)
UCD Moxie	0	556.2	6.9	1.5	0.0	0.0
Cabrillo	0	555.4	7.5	1.3	0.0	0.0
Monterey	0	572.0	9.2	1.2	0.0	0.0
UCD Moxie	7	540.7	7.1	2.6	0.0	0.0
Cabrillo	7	538.7	7.4	2.0	0.2	0.0
Monterey	7	556.3	9.1	1.8	0.0	0.0
UCD Moxie	14	522.0	7.2	3.5	0.0	1.3
Cabrillo	14	522.9	7.3	3.3	0.8	1.2
Monterey	14	542.3	9.5	3.0	0.0	0.5
UCD Moxie	21	506.8	7.0	4.3	9.2	10.3
Cabrillo	21	504.4	7.3	4.3	35.8	36.7
Monterey	21	526.4	8.7	3.8	0.2	11.8

'UCD Moxie' produced fruit meeting or exceeding industry standards for mass-production cultivars (Tables 6-8). The fruit was firm, withstood the rigors of harvest, packing, and storage, and maintained acceptable fruit quality and appearance for over two weeks of storage. 'UCD Moxie' produced significantly firmer fruit than either 'Cabrillo' ($p<0.0001$) or 'Monterey' ($p<0.0001$). 'UCD Moxie' additionally had significantly lower SS concentration than 'Monterey' ($p=0.0049$), but was not significantly different compared to 'Cabrillo' ($p=0.025$). The SS/TA ration for 'UCD Moxie' was significantly lower than 'Monterey' ($p=0.016$) and not significantly different from 'Cabrillo' ($p=0.80$).

'UCD Moxie' maintained adequate marketability and visual appeal over 14 days of post-harvest storage, the industry standard (Table 8), as did 'Cabrillo' and 'Monterey'. The marketability of fruit stored for 21 days post-harvest was inadequate for all three cultivars. Fruit weight and brightness significantly decreased as post-harvest storage time increased (Table 8). The fruit weight decreases were not significantly different among cultivars. Cultivar x post-harvest storage time interactions were only statistically significant for liquid leakage and mold formation, with 'Cabrillo' deteriorating more than 'UCD Moxie' and 'Monterey' (Table 8).

Disease Resistance Evaluation

'UCD Moxie' and additional cultivars were screened for resistance to Fusarium wilt, Verticillium wilt, Macrophomina, and Phytophthora crown rot in Davis, Calif. field experiments between 2015 and 2018. These included 2015-16 and 2016-17 Fusarium wilt screening experiments with 480 to 960 entries, a 2015-16 Macrophomina experiment with 960 entries, 2016-17 and 2017-18 Verticillium wilt experiments with 480 to 960 entries, and a 2017-18 Phytophthora crown rot experiment with 480 entries. Entries were arranged in randomized complete blocks experiment designs with four single-plant replications per entry. The 2015-16 experiments were planted in virgin soil in Davis, Calif. The 2016-17 and 2017-18 experiments were planted in fumigated soils in Davis, Calif. For each experiment, plants were artificially inoculated with the respective pathogen and phenotyped for disease symptoms on an ordinal scale, where 1=highly resistant (symptomless), 2=resistant, 3=intermediate, 4=susceptible, and 5=highly susceptible (dead). Within each experiment, plants were phenotyped at six different time points to study changes in the phenotypic distributions and quantify the progression of disease symptoms over time.

'UCD Moxie' was highly resistant to Fusarium wilt (1.3 on scale), moderately resistant to Verticillium wilt (2.1 on scale), moderately susceptible to Phytophthora crown rot (3.5 on scale), and susceptible to Macrophomina (5.0 on scale).

Botanical Description

The following botanical descriptors are characteristic of 'UCD Moxie'. The descriptors were collected from two different sites in May 2017 in Santa Maria, Calif. Colors are designated with reference to The Royal Horticultural Society (R.H.S.) Colour Chart, Sixth Edition, 2015. The characteristics of 'UCD Moxie' may vary in detail, depending upon environmental factors and culture conditions.

Growth habitat: Semi-upright. Plant height average of 29 cm. Plant spread average of 37 cm.
Density of foliage: Medium.
Vigor: Strong.
Position of inflorescence in relation to foliage: Same level.
Number of stolons: Average of 5.
Stolon, anthocyanin coloration: 60B.
Stolon, density of pubescence: Sparse.
Leaf size: Medium.
Leaf color: Adaxial 137A, Abaxial 147B.
Leaf blistering: Medium.
Leaf glossiness: Medium glossy.
Leaf variegation: Absent.
Terminal leaflet, length in relation to width: Average of 93 mm long and 73 mm wide.
Terminal leaflet, shape of base: Obtuse.
Terminal leaflet, margin: Serrate to crenate.
Terminal leaflet, shape in cross section: Concave.
Petiole, length: Average of 19 cm.
Petiole, attitude of hairs: Slightly outwords.
Stipule, anthocyanin coloration: Core color 144C, Margin color 146A (absent or very weak).
Inflorescence, number of flowers: Many.
Pedicel, attitude of hairs: Slightly outwards.
Pedicel, anthocyanin coloration: 60B.
Flower diameter: Average of 22 mm.
Flower, arrangement of petals: Touching.
Flower, size of calyx: Calyx diameter average of 29 mm.
Color of calyx: 137A.
Flower stamen: Present.
Number of stamens per flower: Average of 23.
Number of sepals per flower: Average of 12.
Petal, length in relation to width: Equal, Average of 10 mm long and 10 mm wide.
Petal, color of upper side: NN155B.
Petal, color of lower side: NN155B.
Number of petals per flower: Average of 6.
Fruit, length in relation to width: Average of 54 mm long and 40 mm wide.
Fruit size: Average of 32 grams through the season for both primary and secondary fruit.
Fruit shape: Conical.
Fruit, difference in shape of terminal and other fruits: None or very slight.
Fruit color: N45A.
Fruit, evenness of color: Even or very slightly uneven.
Fruit glossiness: Strong.
Fruit, evenness of surface: Even or very slightly uneven.
Fruit, width of band without achenes: Absent or very narrow.

Fruit, positions of achenes: Below surface.
Achene color: 4A.
Fruit, position of calyx attachment: Inserted.
Fruit, attitude of sepals: Upwards.
Fruit, diameter of calyx in relation to fruit diameter: Slightly larger.
Fruit, adherence of calyx: Strong.
Fruit firmness: Very firm.
Fruit, color of flesh (excluding core): 40B.
Fruit, color of core: 37B.
Fruit cavity: Average of 3.5 mm.
Time of beginning of flowering: Late, starts in March to December.
Time of beginning of fruit ripening: Late, starts in April to December.
Type of bearing: Day neutral.

Plant and Foliage Comparisons

Fruiting plants of 'UCD Moxie' are similar in height to 'Cabrillo', but taller than 'San Andreas' and slightly shorter than 'Monterey'. The spread is more similar to 'Monterey' and 'Cabrillo' and wider than 'San Andreas'. Leaves (including petioles) for 'UCD Moxie' are similar in length as the three comparative cultivars. Color for the upper and lower levels of the leaves of 'UCD Moxie' are similar to 'Monterey', but darker green than 'Cabrillo' and 'San Andreas'. Serrations at midseason are more pointed than 'Monterey' and more similar in shape and number to 'Cabrillo' and 'San Andreas'. The stipule length of 'UCD Moxie' is longer than all three comparative cultivars. Stolon production of 'UCD Moxie' is similar to 'San Andreas', but less than for 'Cabrillo' and 'Monterey'.

Flowering and Fruiting Comparisons

'UCD Moxie' is similar to other California day-neutral cultivars (e.g. 'Cabrillo', 'Monterey' and 'San Andreas') in that it will flower independently of day length, given appropriate temperature and horticultural conditions. General tendency is for flowering to initiate slightly later than the 3 comparative cultivars, but to maintain exceptionally good strong flowering capacity during the long days of summer. The primary flowers for 'UCD Moxie' are smaller in size to the comparative cultivars with a calyx that is distinctly larger relative to the corolla on the primary fruit. The sepals for 'UCD Moxie' are slightly longer and wider than all 3 comparative cultivars. The calyx of 'UCD Moxie' is mostly reflex similar to 'Cabrillo' and more reflexive than 'Monterey' and 'San Andreas'. The fruit shape of 'UCD Moxie' can vary through the season, but is generally a long conic fruit similar to that of 'San Andreas', comparatively different than the short and rounded conic fruit of 'Cabrillo' and the short and slightly flattened conic fruit of 'Monterey'. External fruit color of 'UCD Moxie' is darker red than both 'Monterey' and 'Cabrillo'. The internal fruit color of 'UCD Moxie' is lighter than both 'Monterey' and 'Cabrillo'.

Achenes of 'UCD Moxie' are even to slightly indented in the fruit, comparatively similar to the fruit of 'Monterey' and less indented than 'Cabrillo' and 'San Andreas'.

What is claimed is:

1. A new and distinct cultivar of strawberry plant having the characteristics substantially as described and illustrated herein.

* * * * *

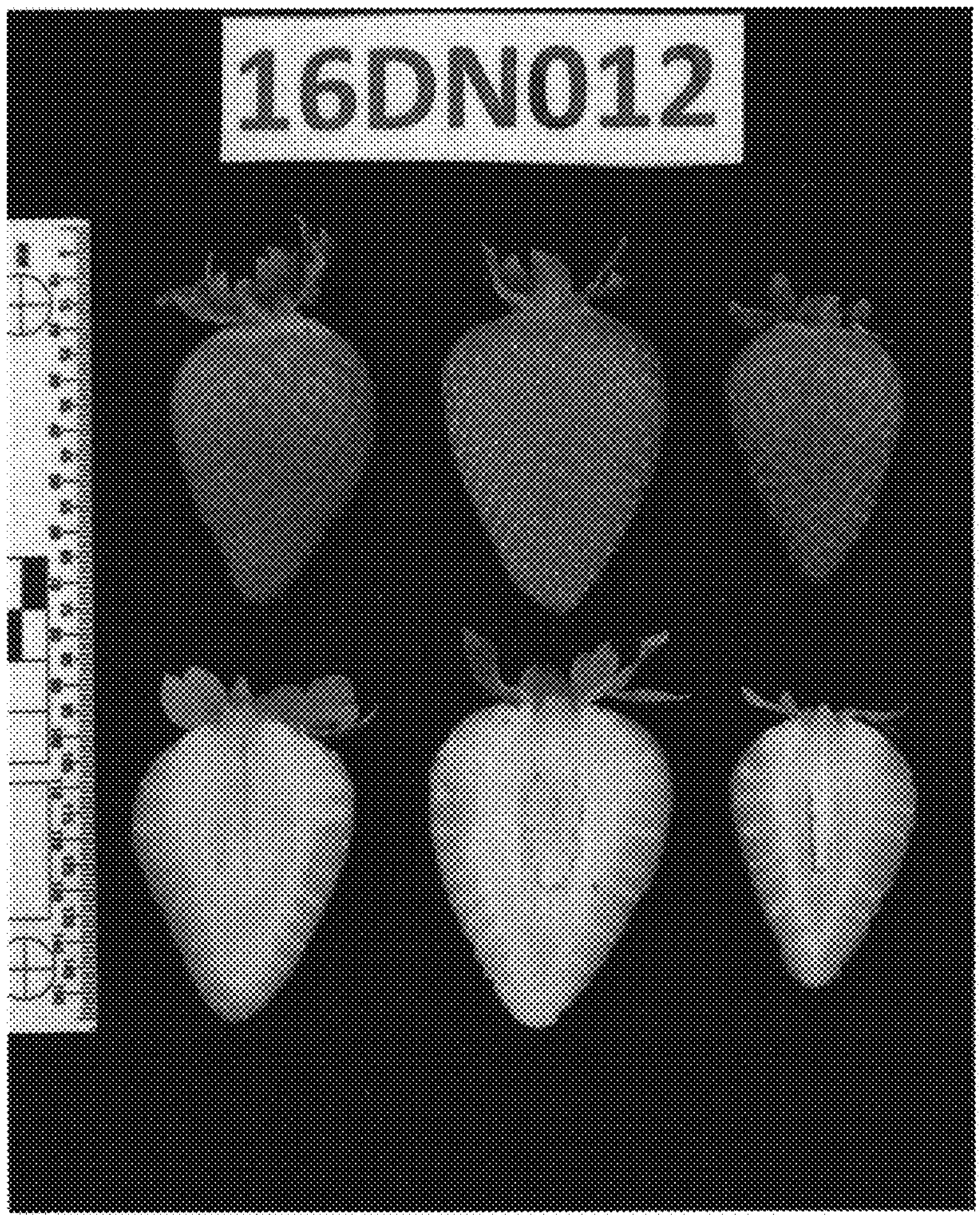

FIG. 1

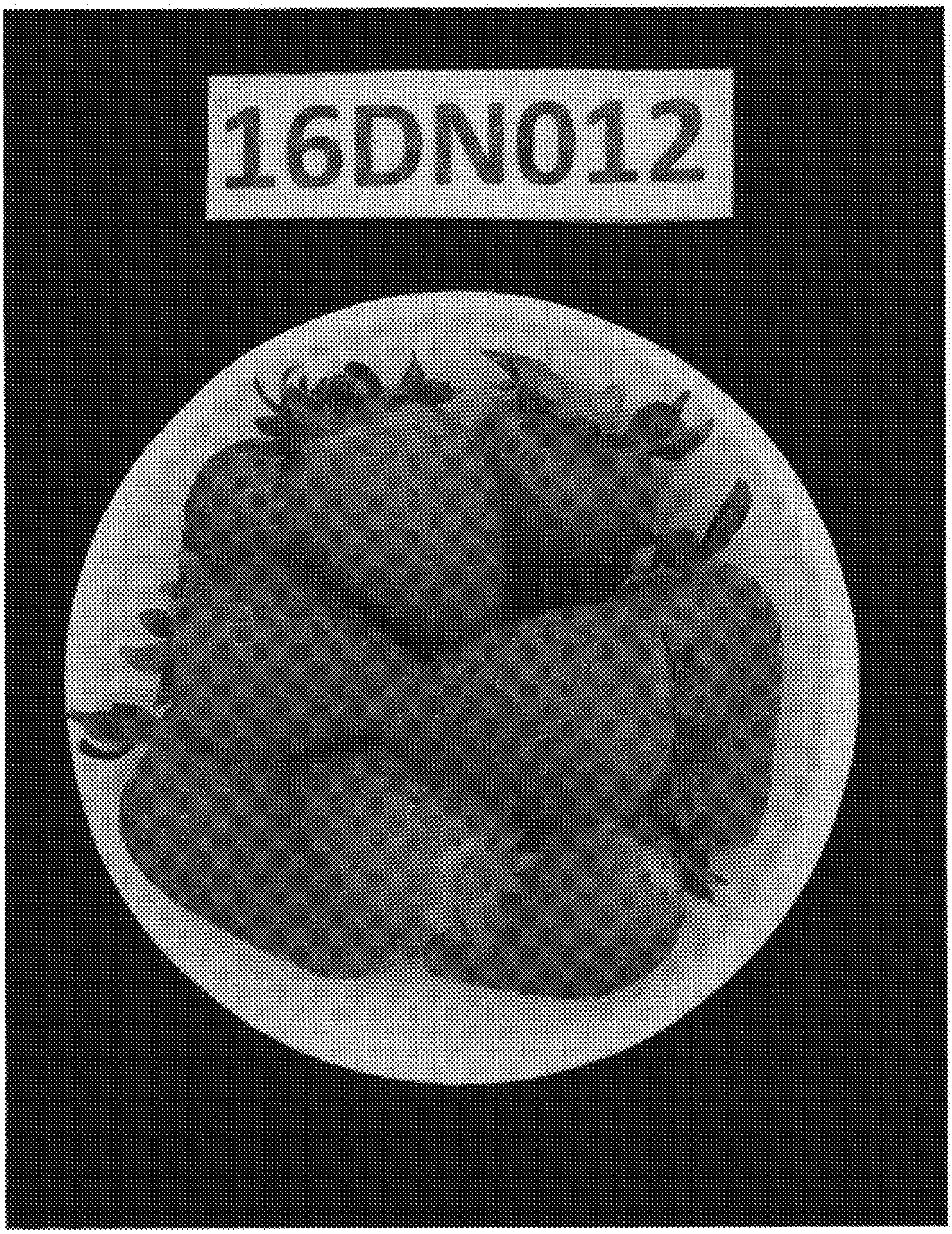

FIG. 2

FIG. 3

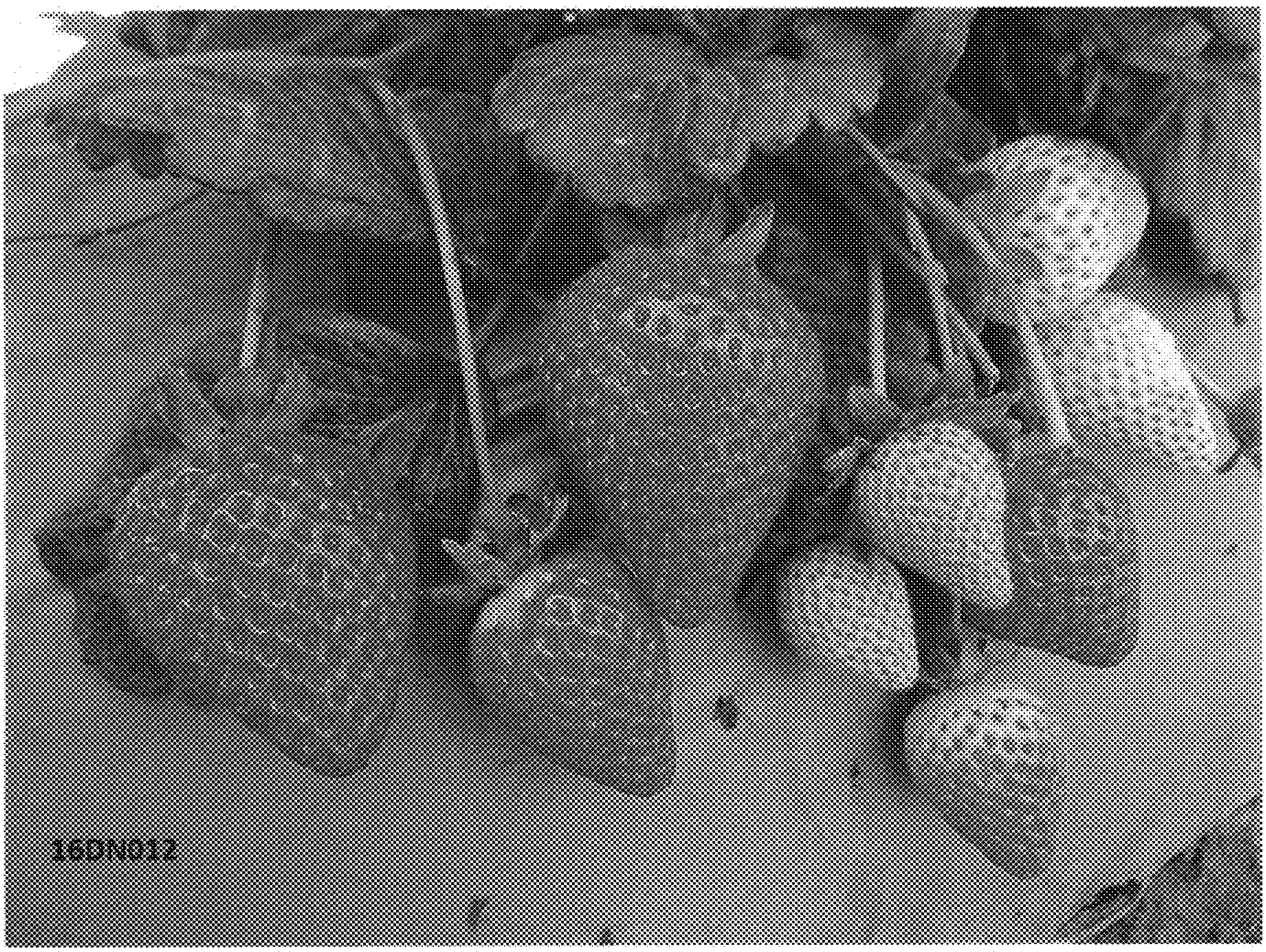

FIG. 4

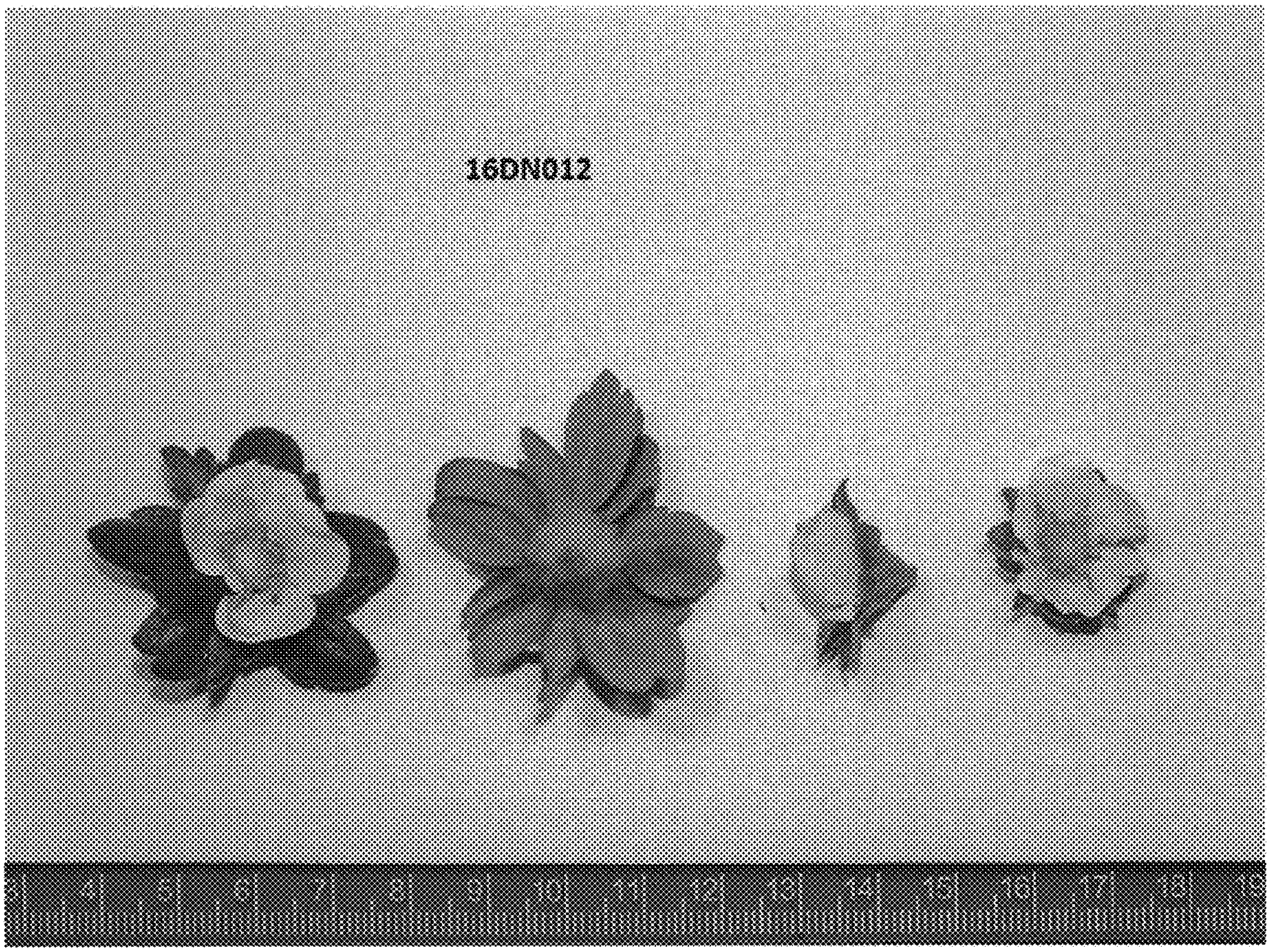

FIG. 5

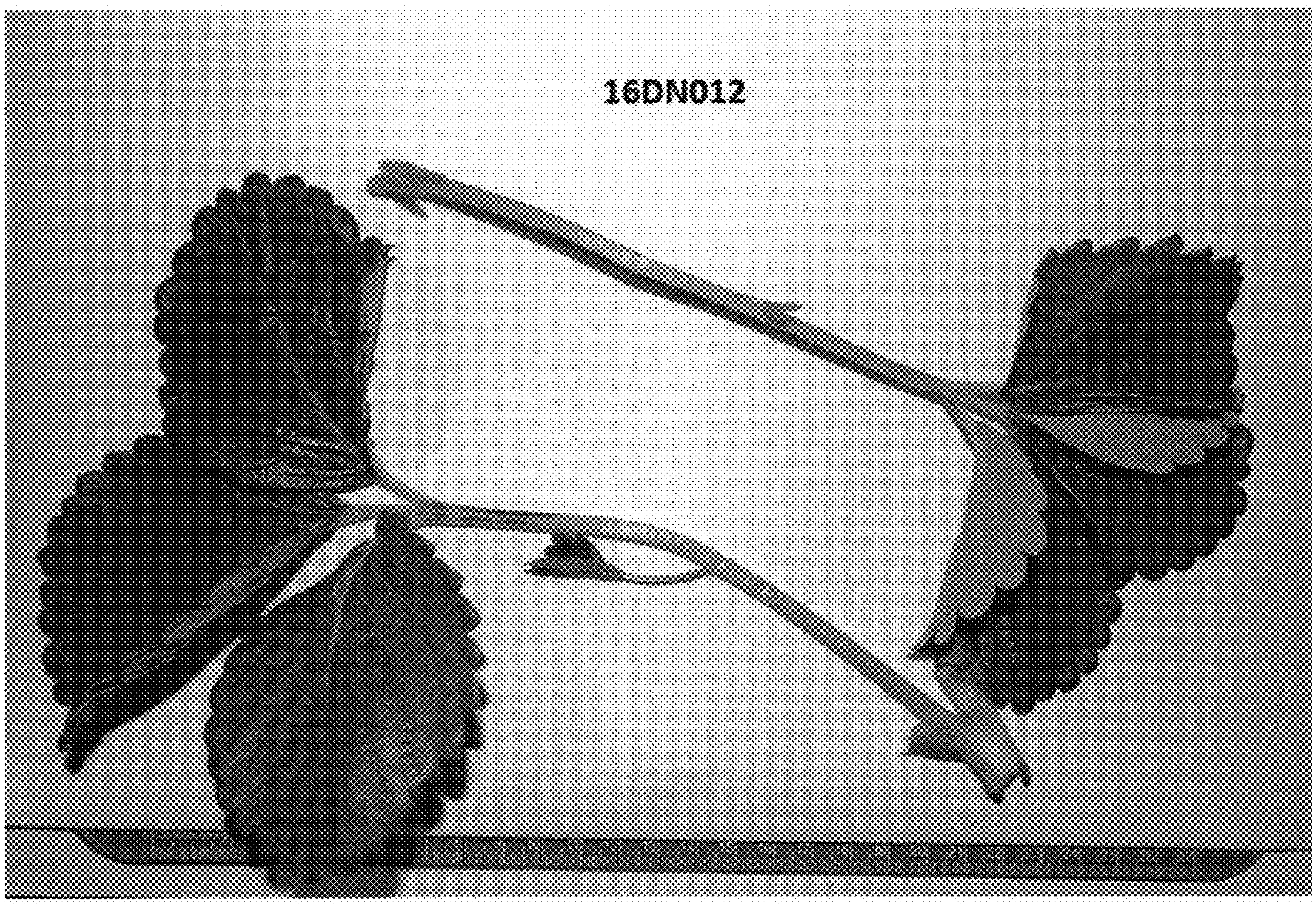

FIG. 6

FIG. 7